Errata

On Graham and Coppock (2021), *Public Opinion Quarterly* 85(1)

This note lists fifteen corrections to "Asking About Attitude Change," published in *Public Opinion Quarterly* 85(1), doi 10.1093/poq/nfab009: one appendix table whose column headings are in the wrong order, two corrections to a single sentence that describes the estimates incorrectly, two misstated numbers, and nine errors in the reference list. **None of them changes a conclusion of the paper.** The evidence behind the paper's argument, that the change format gets the sign of an effect wrong far more often than the counterfactual format does, is unaffected.

Every number in a corrected sentence below, outside the reference list, is computed from the deposited replication archive (10.7910/DVN/GFF78K) at the time this document is rendered, including the numbers that were not wrong. The code and output supporting each correction are at github.com/active-maintenance-aec/graham_coppock_2021.

Every quantity on the percentage scale is stated to one decimal place, and bold marks the corrections. A corrected sentence therefore differs from the published one in more places than it is wrong: the article states these quantities in whole points, and setting a whole number to one decimal is not a correction. Only what is bold has changed.

1. Table D.3's second and third value columns are headed in the wrong order

Table D.3 is headed `More | No change | Less | Difference`. Its body is in the order `More | Less | No change | Difference`, so every reader of the published table who takes the second column for the share reporting no change is reading the share reporting less, and the reverse.

The table's own Difference column is what settles which side is wrong. In 84 of its 84 rows the printed Difference equals More minus the column headed "No change", to within the one unit in the last printed digit that rounding each cell separately allows. It equals More minus the column headed "Less" in 2 of them. Since the difference the article defines is the share reporting more minus the share reporting less, the column headed "No change" holds the share reporting less. The main text reads the body the same way: of the Study 1 disputed accusation row for Democrats it says that "a huge majority of Democratic respondents (87 percent) report that the accusation made them less likely to vote for Cornish," and 86.8 is

the entry under "No change"; of the Republican row it says "most Republicans (57 percent) reported that the information had no effect," and 56.8 is the entry under "Less".

The table is what is corrected, not any sentence. Every cell in the published body is right and sits in the right place; the two headings over the second and third value columns are transposed. Table D.3 is reprinted below with the headings in the order the body uses. The numbers are unchanged: they are recomputed from the deposited data and reproduce the published table cell for cell.

Table 1: Table D.3 corrected: Distribution of Change Format by Study and Treatment. Standard errors in parentheses. Only the headings over the third and fourth value columns differ from the published table.

Treatment	Party	Level Q?	N	More	Less	No change	Difference
Blocked whistleblower (Study 1)	Democrat	No	53	22.6 (5.8)	45.3 (6.9)	32.1 (6.5)	-22.6 (11.0)
Blocked whistleblower (Study 1)	Democrat	Yes	46	13.0 (5.0)	50.0 (7.5)	37.0 (7.2)	-37.0 (10.5)
Blocked whistleblower (Study 1)	Independent	No	14	14.3 (9.7)	21.4 (11.4)	64.3 (13.3)	-7.1 (16.5)
Blocked whistleblower (Study 1)	Independent	Yes	10	0.0 (0.0)	30.0 (15.3)	70.0 (15.3)	-30.0 (15.3)
Blocked whistleblower (Study 1)	Republican	No	37	2.7 (2.7)	59.5 (8.2)	37.8 (8.1)	-56.8 (9.1)
Blocked whistleblower (Study 1)	Republican	Yes	47	6.4 (3.6)	38.3 (7.2)	55.3 (7.3)	-31.9 (8.7)
Death penalty (Study 1)	Democrat	No	52	9.6 (4.1)	34.6 (6.7)	55.8 (7.0)	-25.0 (8.6)
Death penalty (Study 1)	Democrat	Yes	45	8.9 (4.3)	20.0 (6.0)	71.1 (6.8)	-11.1 (7.9)
Death penalty (Study 1)	Independent	No	12	16.7 (11.2)	0.0 (0.0)	83.3 (11.2)	16.7 (11.2)
Death penalty (Study 1)	Independent	Yes	10	40.0 (16.3)	10.0 (10.0)	50.0 (16.7)	30.0 (21.3)
Death penalty (Study 1)	Republican	No	37	24.3 (7.2)	8.1 (4.5)	67.6 (7.8)	16.2 (9.1)
Death penalty (Study 1)	Republican	Yes	47	27.7 (6.6)	8.5 (4.1)	63.8 (7.1)	19.1 (8.4)
Disputed accusation (Study 1)	Democrat	No	53	1.9 (1.9)	86.8 (4.7)	11.3 (4.4)	-84.9 (5.6)
Disputed accusation (Study 1)	Democrat	Yes	46	2.2 (2.2)	63.0 (7.2)	34.8 (7.1)	-60.9 (7.9)
Disputed accusation (Study 1)	Independent	No	13	7.7 (7.7)	46.2 (14.4)	46.2 (14.4)	-38.5 (18.0)
Disputed accusation (Study 1)	Independent	Yes	10	10.0 (10.0)	50.0 (16.7)	40.0 (16.3)	-40.0 (22.1)
Disputed accusation (Study 1)	Republican	No	37	18.9 (6.5)	24.3 (7.2)	56.8 (8.3)	-5.4 (10.9)
Disputed accusation (Study 1)	Republican	Yes	47	8.5 (4.1)	38.3 (7.2)	53.2 (7.4)	-29.8 (9.1)
Endorsed Trump (Study 1)	Democrat	No	53	5.7 (3.2)	79.2 (5.6)	15.1 (5.0)	-73.6 (7.7)
Endorsed Trump (Study 1)	Democrat	Yes	46	2.2 (2.2)	76.1 (6.4)	21.7 (6.1)	-73.9 (7.2)
Endorsed Trump (Study 1)	Independent	No	14	0.0 (0.0)	28.6 (12.5)	71.4 (12.5)	-28.6 (12.5)
Endorsed Trump (Study 1)	Independent	Yes	10	20.0 (13.3)	40.0 (16.3)	40.0 (16.3)	-20.0 (24.9)
Endorsed Trump (Study 1)	Republican	No	37	59.5 (8.2)	5.4 (3.8)	35.1 (8.0)	54.1 (10.0)
Endorsed Trump (Study 1)	Republican	Yes	47	31.9 (6.9)	23.4 (6.2)	44.7 (7.3)	8.5 (10.9)
Immigration (Study 1)	Democrat	No	52	23.1 (5.9)	17.3 (5.3)	59.6 (6.9)	5.8 (8.9)
Immigration (Study 1)	Democrat	Yes	45	17.8 (5.8)	11.1 (4.7)	71.1 (6.8)	6.7 (8.0)
Immigration (Study 1)	Independent	No	12	16.7 (11.2)	8.3 (8.3)	75.0 (13.1)	8.3 (14.9)
Immigration (Study 1)	Independent	Yes	10	20.0 (13.3)	20.0 (13.3)	60.0 (16.3)	0.0 (21.1)
Immigration (Study 1)	Republican	No	37	10.8 (5.2)	45.9 (8.3)	43.2 (8.3)	-35.1 (11.1)
Immigration (Study 1)	Republican	Yes	47	14.9 (5.2)	29.8 (6.7)	55.3 (7.3)	-14.9 (9.6)
Supports charters (Study 1)	Democrat	No	53	20.8 (5.6)	43.4 (6.9)	35.8 (6.7)	-22.6 (10.7)
Supports charters (Study 1)	Democrat	Yes	46	17.4 (5.7)	39.1 (7.3)	43.5 (7.4)	-21.7 (10.7)

Table 1: Table D.3 corrected: Distribution of Change Format by Study and Treatment. Standard errors in parentheses. Only the headings over the third and fourth value columns differ from the published table. *(continued)*

Treatment	Party	Level Q?	N	More	Less	No change	Difference
Supports charters (Study 1)	Independent	No	14	21.4 (11.4)	14.3 (9.7)	64.3 (13.3)	7.1 (16.5)
Supports charters (Study 1)	Independent	Yes	10	20.0 (13.3)	10.0 (10.0)	70.0 (15.3)	10.0 (18.0)
Supports charters (Study 1)	Republican	No	37	54.1 (8.3)	13.5 (5.7)	32.4 (7.8)	40.5 (11.9)
Supports charters (Study 1)	Republican	Yes	47	40.4 (7.2)	12.8 (4.9)	46.8 (7.4)	27.7 (9.9)
Tax cuts (Study 1)	Democrat	No	52	11.5 (4.5)	57.7 (6.9)	30.8 (6.5)	-46.2 (9.7)
Tax cuts (Study 1)	Democrat	Yes	45	8.9 (4.3)	33.3 (7.1)	57.8 (7.4)	-24.4 (9.1)
Tax cuts (Study 1)	Independent	No	12	33.3 (14.2)	16.7 (11.2)	50.0 (15.1)	16.7 (20.7)
Tax cuts (Study 1)	Independent	Yes	10	30.0 (15.3)	30.0 (15.3)	40.0 (16.3)	0.0 (25.8)
Tax cuts (Study 1)	Republican	No	37	70.3 (7.6)	0.0 (0.0)	29.7 (7.6)	70.3 (7.6)
Tax cuts (Study 1)	Republican	Yes	47	29.8 (6.7)	10.6 (4.5)	59.6 (7.2)	19.1 (8.9)
Undisputed accusation (Study 1)	Democrat	No	53	1.9 (1.9)	81.1 (5.4)	17.0 (5.2)	-79.2 (6.2)
Undisputed accusation (Study 1)	Democrat	Yes	45	2.2 (2.2)	68.9 (7.0)	28.9 (6.8)	-66.7 (7.8)
Undisputed accusation (Study 1)	Independent	No	13	15.4 (10.4)	53.8 (14.4)	30.8 (13.3)	-38.5 (21.3)
Undisputed accusation (Study 1)	Independent	Yes	10	0.0 (0.0)	80.0 (13.3)	20.0 (13.3)	-80.0 (13.3)
Undisputed accusation (Study 1)	Republican	No	37	5.4 (3.8)	73.0 (7.4)	21.6 (6.9)	-67.6 (9.5)
Undisputed accusation (Study 1)	Republican	Yes	47	8.5 (4.1)	61.7 (7.2)	29.8 (6.7)	-53.2 (9.5)
Biden / Hill (Study 2a)	Democrat	No	332	12.7 (1.8)	24.4 (2.4)	63.0 (2.7)	-11.7 (3.3)
Biden / Hill (Study 2a)	Independent	No	135	3.0 (1.5)	14.1 (3.0)	83.0 (3.2)	-11.1 (3.4)
Biden / Hill (Study 2a)	Republican	No	355	8.2 (1.5)	22.5 (2.2)	69.3 (2.5)	-14.4 (2.8)
DREAM Act (Study 2a)	Democrat	No	332	50.0 (2.7)	7.8 (1.5)	42.2 (2.7)	42.2 (3.5)
DREAM Act (Study 2a)	Independent	No	135	29.6 (3.9)	10.4 (2.6)	60.0 (4.2)	19.3 (5.2)
DREAM Act (Study 2a)	Republican	No	355	23.1 (2.2)	26.2 (2.3)	50.7 (2.7)	-3.1 (3.7)
Obama Torture (Study 2a)	Democrat	No	332	43.1 (2.7)	7.5 (1.5)	49.4 (2.7)	35.5 (3.4)
Obama Torture (Study 2a)	Independent	No	135	23.0 (3.6)	11.9 (2.8)	65.2 (4.1)	11.1 (5.0)
Obama Torture (Study 2a)	Republican	No	356	18.3 (2.1)	18.0 (2.0)	63.8 (2.6)	0.3 (3.2)
Opposed Kav (Study 2a)	Democrat	No	237	56.5 (3.2)	11.4 (2.1)	32.1 (3.0)	45.1 (4.5)
Opposed Kav (Study 2a)	Independent	No	96	22.9 (4.3)	15.6 (3.7)	61.5 (5.0)	7.3 (6.3)
Opposed Kav (Study 2a)	Republican	No	225	9.8 (2.0)	59.6 (3.3)	30.7 (3.1)	-49.8 (4.5)
Supported Kav (Study 2a)	Democrat	No	95	3.2 (1.8)	71.6 (4.7)	25.3 (4.5)	-68.4 (5.5)
Supported Kav (Study 2a)	Independent	No	39	7.7 (4.3)	25.6 (7.1)	66.7 (7.6)	-17.9 (8.9)
Supported Kav (Study 2a)	Republican	No	130	51.5 (4.4)	10.8 (2.7)	37.7 (4.3)	40.8 (6.0)
Trump Coal (Study 2a)	Democrat	No	332	30.1 (2.5)	29.2 (2.5)	40.7 (2.7)	0.9 (4.2)
Trump Coal (Study 2a)	Independent	No	135	19.3 (3.4)	15.6 (3.1)	65.2 (4.1)	3.7 (5.1)
Trump Coal (Study 2a)	Republican	No	356	25.8 (2.3)	11.0 (1.7)	63.2 (2.6)	14.9 (3.1)
Biden / Hill (Study 2b)	Democrat	No	132	9.1 (2.5)	17.4 (3.3)	73.5 (3.9)	-8.3 (4.4)
Biden / Hill (Study 2b)	Independent	No	49	0.0 (0.0)	20.4 (5.8)	79.6 (5.8)	-20.4 (5.8)
Biden / Hill (Study 2b)	Republican	No	108	11.1 (3.0)	25.0 (4.2)	63.9 (4.6)	-13.9 (5.7)
DREAM Act (Study 2b)	Democrat	No	131	54.2 (4.4)	6.1 (2.1)	39.7 (4.3)	48.1 (5.3)
DREAM Act (Study 2b)	Independent	No	49	22.4 (6.0)	18.4 (5.6)	59.2 (7.1)	4.1 (9.2)
DREAM Act (Study 2b)	Republican	No	108	29.6 (4.4)	34.3 (4.6)	36.1 (4.6)	-4.6 (7.7)
Obama Torture (Study 2b)	Democrat	No	131	55.0 (4.4)	9.2 (2.5)	35.9 (4.2)	45.8 (5.8)
Obama Torture (Study 2b)	Independent	No	49	26.5 (6.4)	16.3 (5.3)	57.1 (7.1)	10.2 (9.3)
Obama Torture (Study 2b)	Republican	No	108	25.0 (4.2)	34.3 (4.6)	40.7 (4.8)	-9.3 (7.4)
Trump Coal (Study 2b)	Democrat	No	131	32.8 (4.1)	36.6 (4.2)	30.5 (4.0)	-3.8 (7.3)
Trump Coal (Study 2b)	Independent	No	49	16.3 (5.3)	26.5 (6.4)	57.1 (7.1)	-10.2 (9.3)
Trump Coal (Study 2b)	Republican	No	108	32.4 (4.5)	16.7 (3.6)	50.9 (4.8)	15.7 (6.6)
Mueller comments (Study 3)	Democrat	No	256	58.2 (3.4)	10.2 (1.9)	31.6 (3.1)	48.0 (4.5)
Mueller comments (Study 3)	Democrat	Yes	240	42.1 (3.4)	12.9 (2.2)	45.0 (3.4)	29.2 (4.6)

Table 1: Table D.3 corrected: Distribution of Change Format by Study and Treatment. Standard errors in parentheses. Only the headings over the third and fourth value columns differ from the published table. *(continued)*

Treatment	Party	Level Q?	N	More	Less	No change	Difference
Mueller comments (Study 3)	Independent	No	131	33.6 (4.3)	16.8 (3.5)	49.6 (4.6)	16.8 (6.4)
Mueller comments (Study 3)	Independent	Yes	114	23.7 (4.0)	14.0 (3.3)	62.3 (4.6)	9.6 (5.7)
Mueller comments (Study 3)	Republican	No	168	12.5 (2.9)	33.3 (4.0)	54.2 (4.2)	-20.8 (5.5)
Mueller comments (Study 3)	Republican	Yes	165	13.3 (3.0)	27.9 (3.7)	58.8 (4.1)	-14.5 (5.3)

2. The counterfactual format's sign errors are not all cases in which nothing can be distinguished from zero

> The more-minus-less estimates derived from the change format have the opposite sign as the experimental difference-in-means estimates in 12 out of 20 opportunities, compared with 3 of 20 for the counterfactual format. **In all three of those cases**, neither the counterfactual estimate nor the difference in means estimate can be distinguished from zero.

Corrected:

> The more-minus-less estimates derived from the change format have the opposite sign as the experimental difference-in-means estimates in 12 out of 20 opportunities, compared with 2 of 20 for the counterfactual format. **In one of those two cases**, neither the counterfactual estimate nor the difference in means estimate can be distinguished from zero.

The exception is the disputed accusation among Democrats, where the counterfactual estimate is –0.49 scale points with a 95 per cent interval of (–0.71, –0.27), which excludes zero. The article's own Table D.5 prints –0.49 with an interval of (–0.71, –0.28) for the same cell, so the sentence disagrees with the appendix as well as with the data. The other half of the sentence holds there: the difference in means for that cell is 0.02 with an interval of (–0.61, 0.65). Figure 4 and Table D.5 both carry the correct estimates and are not reprinted. How many cells the sentence should count is a separate correction, entry 5 below, which removes a different cell from the three and leaves this one standing.

What the sentence was there to establish still holds in a narrower form. The counterfactual estimate excludes zero in this cell, but the experiment is imprecise (standard error 0.31), and the difference between the two estimates, 0.51, has a bootstrap interval of (–0.07, 1.09). The cell is not among the six differences the article reports as statistically significant, and the other sign error, supports charters among Democrats, is –0.06 against 0.01 with an interval of (–0.63, 0.78) for their difference. Neither of the counterfactual format's sign errors is a statistically distinguishable disagreement with the experiment.

3. The effect of asking the level question first among Democrats

> The effect is larger among Democrats (**14** points, SE: 3 points) and Republicans (8 points, SE: 4 points) than among pure independents (2 points, SE: 6 points).

Corrected:

> The effect is larger among Democrats (**13.5** points, SE: 3.2 points) and Republicans (7.9 points, SE: 4.3 points) than among pure independents (2.2 points, SE: 6.2 points).

The Democratic estimate is 13.4588 points, which rounds to 13 rather than 14 at the whole-point precision the article uses. The comparison the sentence makes, that the effect is larger among Democrats and Republicans than among pure independents, holds.

4. The correlation between the two impeachment questions

> Panel (a) offers prima facie evidence that subjects are not bewildered by the nonstandard question format: the strong correlation (**0.82**) of the two responses indicates that most subjects believe their counterfactual attitudes would have been very close to their actual attitudes (72 percent report exactly zero change).

Corrected:

> Panel (a) offers prima facie evidence that subjects are not bewildered by the nonstandard question format: the strong correlation (**0.83**) of the two responses indicates that most subjects believe their counterfactual attitudes would have been very close to their actual attitudes (72.1 percent report exactly zero change).

The correlation is 0.8285, which rounds up rather than down at two decimals. The share reporting exactly zero change is 72.0873 per cent and is correct as published at whole-percent precision.

5. The counterfactual format has two sign errors, not three

> The more-minus-less estimates derived from the change format have the opposite sign as the experimental difference-in-means estimates in 12 out of 20 opportunities, compared with **3 of 20** for the counterfactual format. In all three of those cases, neither the counterfactual estimate nor the difference in means estimate can be distinguished from zero.

Corrected:

> The more-minus-less estimates derived from the change format have the opposite sign as the experimental difference-in-means estimates in 12 out of 20 opportunities, compared with **2 of 20** for the counterfactual format. In one of those two cases, neither the counterfactual estimate nor the difference in means estimate can be distinguished from zero.

One of the three cells the sentence counts, the Trump coal ash executive order among Republicans, has a counterfactual estimate of 0.00 scale points (SE 0.05). An estimate of zero has no sign, so it cannot have the opposite sign of the experiment, and it is not counted here. The Tax Cuts and Jobs Act cell among Republicans is zero in the same way, differing from zero by 1e-17 scale points, and is also not counted.

Comparing the two signs directly, which is what produces the published count, makes the answer turn on the last bits of that second value. The same code over the same data returns 3 under the linear algebra the published archive was run against and 4 under a current one, so the published count is not reproducible in either direction rather than simply being one too high.

The direction of the article's argument is untouched, and the corrected numbers are wider apart than the published ones: the change format gets the sign wrong in 12 of 20 opportunities against the counterfactual format's 2.

6. Page (2009) is printed with a broken URL

> story?id=8316019**&page**=1.

Corrected:

> story?id=8316019**&page**=1.

The URL is printed with a backslash before its ampersand, a LaTeX escape that reached the page. The story (Susan Page, USA Today, August 13, 2009) is at https://abcnews.go.com/amp/Politics/story?id=8316019&page=1.

7. Guess and Coppock (2020) is given the page of its corrigendum

> British Journal of Political Science 50: **1517**.

Corrected:

> British Journal of Political Science 50: **1497–1515**.

Crossref, doi 10.1017/S0007123418000327: the article runs from p. 1497 to p. 1515 of volume 50, issue 4. Page 1517 is its corrigendum (doi 10.1017/S0007123419000024).

8. Jerzy Neyman's given name is misspelled

> Neyman, **Jerzey**. 1923.

Corrected:

> Neyman, **Jerzy**. 1923.

The author is Jerzy Neyman (Jerzy Splawa-Neyman in the 1990 translation, Statistical Science 5(4), doi 10.1214/ss/1177012031).

9. Strack and Martin (1987) prints its editors' names inverted

> Survey Methodology, edited by **J. Hippler Hans, Schwarz Norbert, and Sudman Seymour**, 123–48.

Corrected:

> Survey Methodology, edited by **Hans-J. Hippler, Norbert Schwarz, and Seymour Sudman**, 123–48.

Crossref, doi 10.1007/978-1-4612-4798-2: the volume is edited by Hans-J. Hippler, Norbert Schwarz, and Seymour Sudman. The entry prints each given name after its surname, as if it were one.

10. Strack and Martin (1987) misquotes its own title

> Account of Context Effects in **Attitudes** Surveys.

Corrected:

> Account of Context Effects in **Attitude** Surveys.

Crossref, doi 10.1007/978-1-4612-4798-2_7: the chapter is "Thinking, Judging, and Communicating: A Process Account of Context Effects in Attitude Surveys."

11. Briñol and Petty (2012) prints an editor's name inverted

> Handbook of Self-Knowledge, edited by **Vazire Simine** and Timothy D. Wilson,

Corrected:

> Handbook of Self-Knowledge, edited by **Simine Vazire** and Timothy D. Wilson,

The Handbook of Self-Knowledge (Guilford Press, 2012) is edited by Simine Vazire and Timothy D. Wilson, as the chapter's own header and the publisher's contents page give it.

12. Schryer and Ross (2012) gives its editors in the wrong order

In Social Metacognition, eds. **Kenneth G. DeMarree, Pablo Briñol**, 141–57.

Corrected:

In Social Metacognition, eds. **Pablo Briñol and Kenneth G. DeMarree**, 141–57.

Crossref, doi 10.4324/9780203865989: Social Metacognition (Psychology Press, 2012) is edited by Pablo Briñol and Kenneth G. DeMarree, in that order.

13. Tourangeau, Rips, and Rasinski (2000) gives the first author the wrong given name

Tourangeau, **Robert**, Lance J. Rips,

Corrected:

Tourangeau, **Roger**, Lance J. Rips,

Crossref, doi 10.1017/CBO9780511819322: the first author is Roger Tourangeau.

14. Yair and Huber (2021) misquotes its own title

How Robust Is Evidence of **Perceptual Partisan** Bias in Survey Responses?

Corrected:

How Robust Is Evidence of **Partisan Perceptual** Bias in Survey Responses?

Public Opinion Quarterly 84(2), doi 10.1093/poq/nfaa024: the title is "How Robust Is Evidence of Partisan Perceptual Bias in Survey Responses? A New Approach for Studying Expressive Responding."

15. Yeh et al. (2018) omits its article number

British Medical Journal **363**.

Corrected:

British Medical Journal **363: k5094**.

Crossref, doi 10.1136/bmj.k5094: the article is BMJ 363: k5094. The BMJ paginates by article number, so without it the reference has no locator.